Lecture 3: Data Visualization

Lecturer

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CLASS CODE

TOAE203

COURSE

Statistics and Probability in Economics and Business

Five blocks, one question running through all of them

What does a summary statistic throw away, and how do we get it back?

01

WHY PLOT AT ALL

Four datasets with identical statistics that are not remotely the same data

02

THE GRAMMAR OF A CHART

A chart maps numbers onto a physical channel, and your eye reads each one differently

03

DIFFERENT TYPE OF CHARTS

You build the four figures that Sessions 5, 7 and 13 will reopen

04

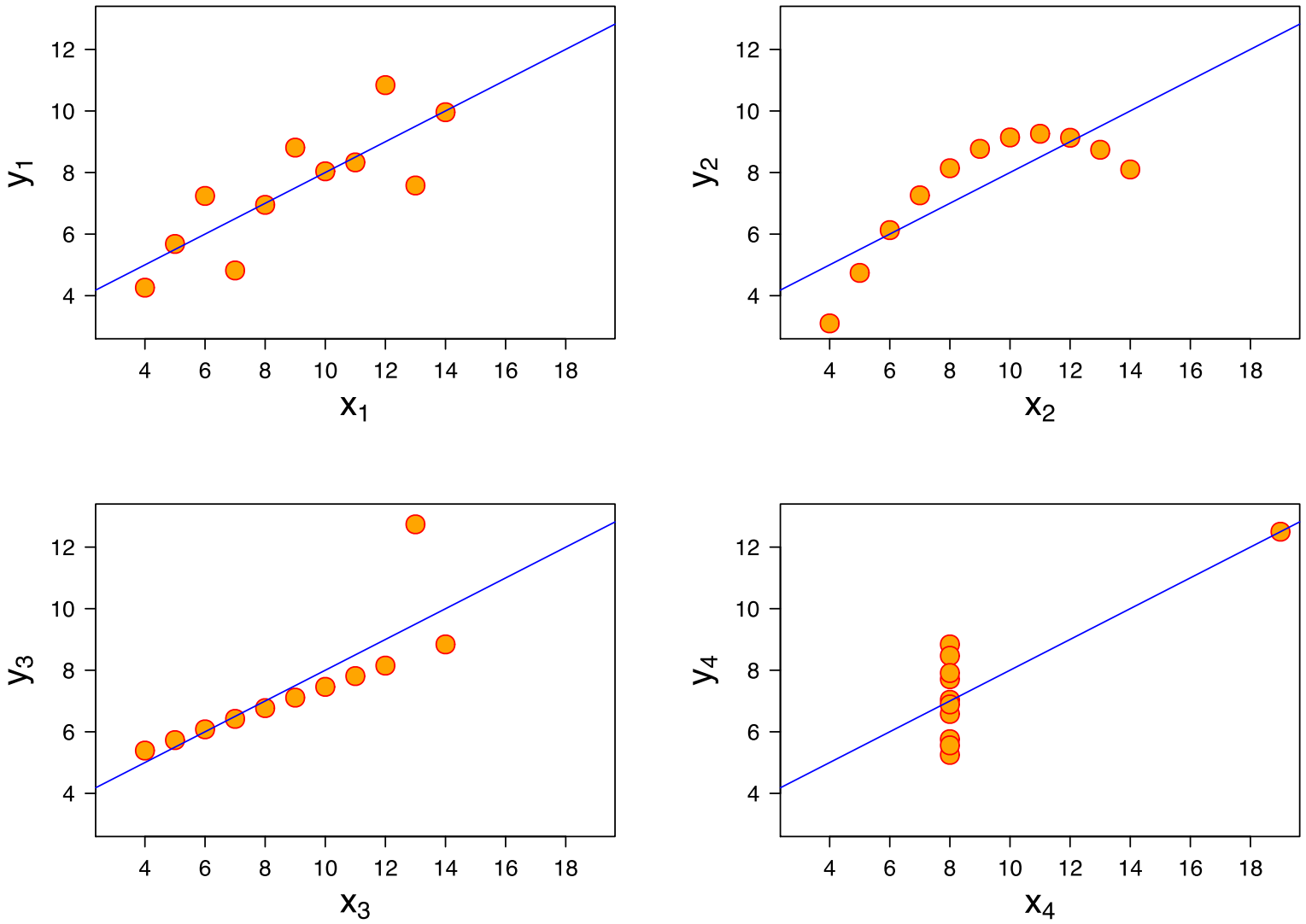
WORKSHOP — DIAGNOSE & PRESCRIBE

Six published charts. Every number in them is correct. Find the encoding decision

WHY PLOT AT ALL

Four datasets. Are they the same dataset?

Anscombe's quartet — eleven points each, four separate files



WHY PLOT AT ALL

Four datasets. Are they the same dataset?

Anscombe's quartet — eleven points each, four separate files

Statistic	Set I = Set II = Set III = Set IV
Mean of x	9.00
Sample variance of x	11.00
Mean of y	7.50
Sample variance of y	4.13
Correlation of x and y	0.816
Fitted line	y = 3.00 + 0.500x
R squared	0.67

Hands up before we go on — same data, or different data?

WHY PLOT AT ALL

The same seven metrics. Four entirely different worlds

ENCODING

A summary is a weighted sum over every row

Mean, variance and correlation are each a sum across all rows. A sum has no memory of which row contributed what. Move one point up and another down by matching amounts, and the sum does not move.

READING

What the eye recovers in two seconds

- Set I — a real linear relationship with scatter around it
- Set II — a clean curve; the straight line is the wrong shape
- Set III — a perfect line plus one outlier dragging the slope
- Set IV — no relationship at all; one point at x = 19 creates it

A statistic answers the question you thought to ask. A picture shows the question you did not.

WHY PLOT AT ALL

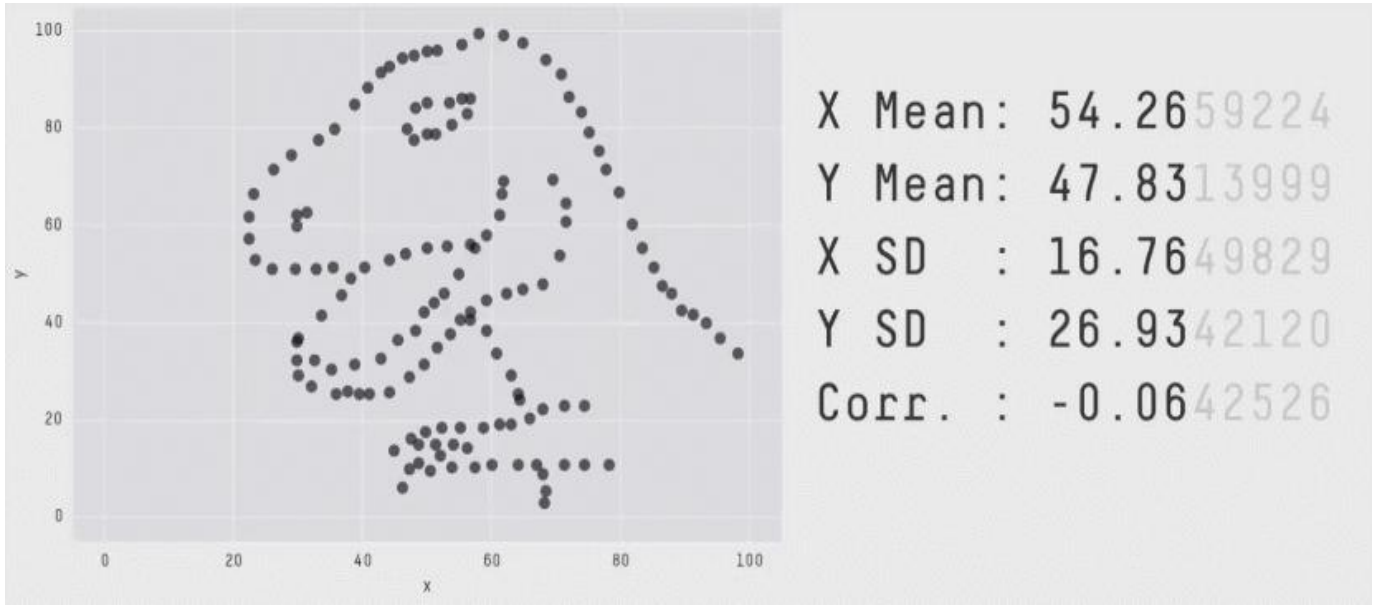
The Datasaurus

Thirteen datasets sharing the same summary statistics. One of them is a dinosaur.

ENCODING

Shape can be chosen freely underneath the statistics

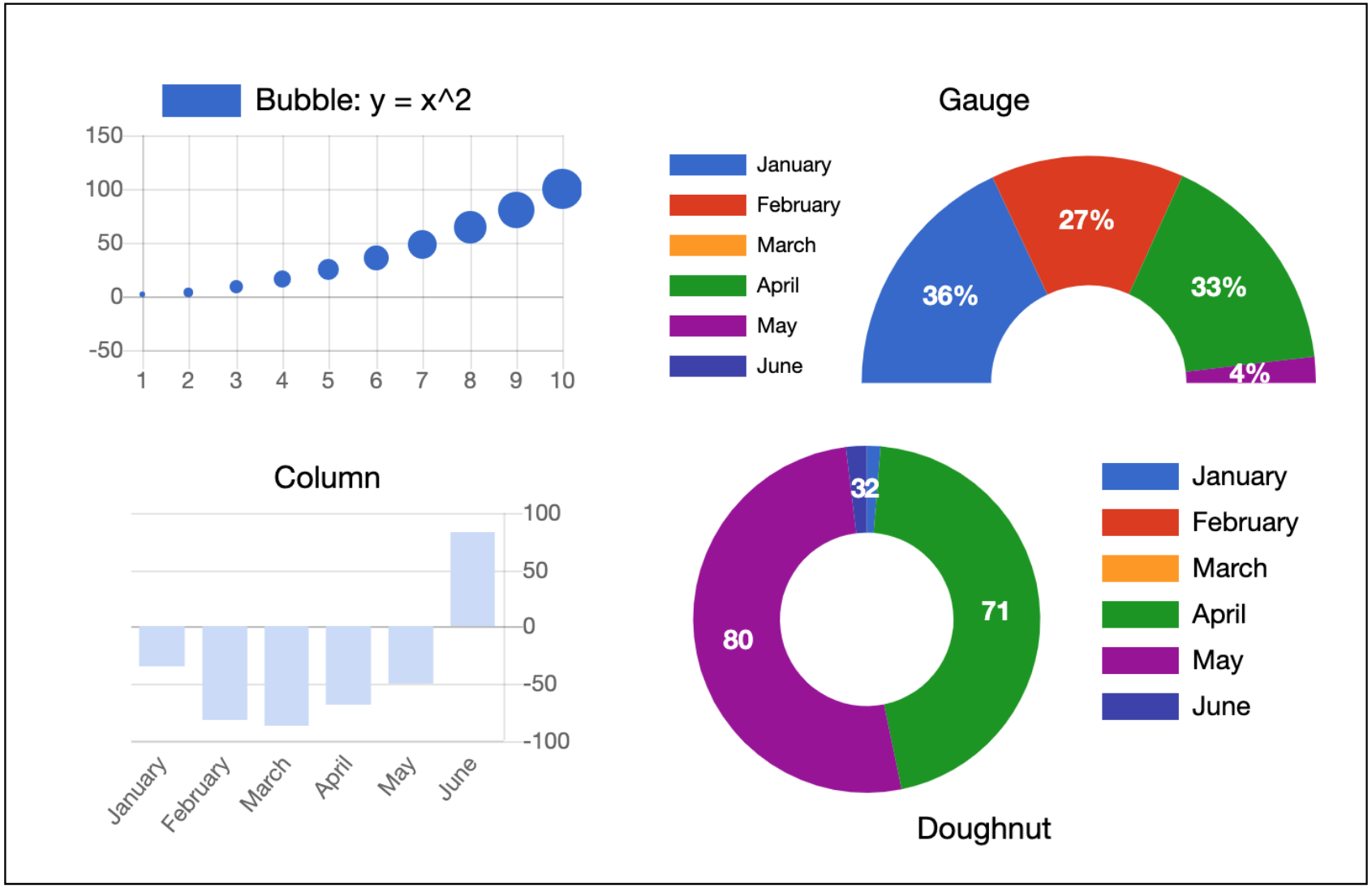
Matejka and Fitzmaurice showed you can nudge points one small step at a time into almost any shape you like, while holding the means, the standard deviations and the correlation fixed to two decimal places.



Never report a statistic for a variable whose picture you have not looked at.

Components of a Chart

A chart turns a number into something your eye then has to measure back



THE GRAMMAR OF A CHART

Components of a Chart

A chart turns a number into something your eye then has to measure back

01

POSITION on a shared axis

Bar height, dot location, line vertex. Read almost perfectly.

02

LENGTH, with no shared baseline

Floating bars, stacked segments. Read well, but not as well.

03

ANGLE or SLOPE

Pie wedges, line steepness. Read poorly and inconsistently.

04

AREA

Bubbles, treemap tiles, 3D column faces. Large values underestimated.

05

COLOUR and SHADING

Hue for category, saturation for quantity. The weakest channel of all.

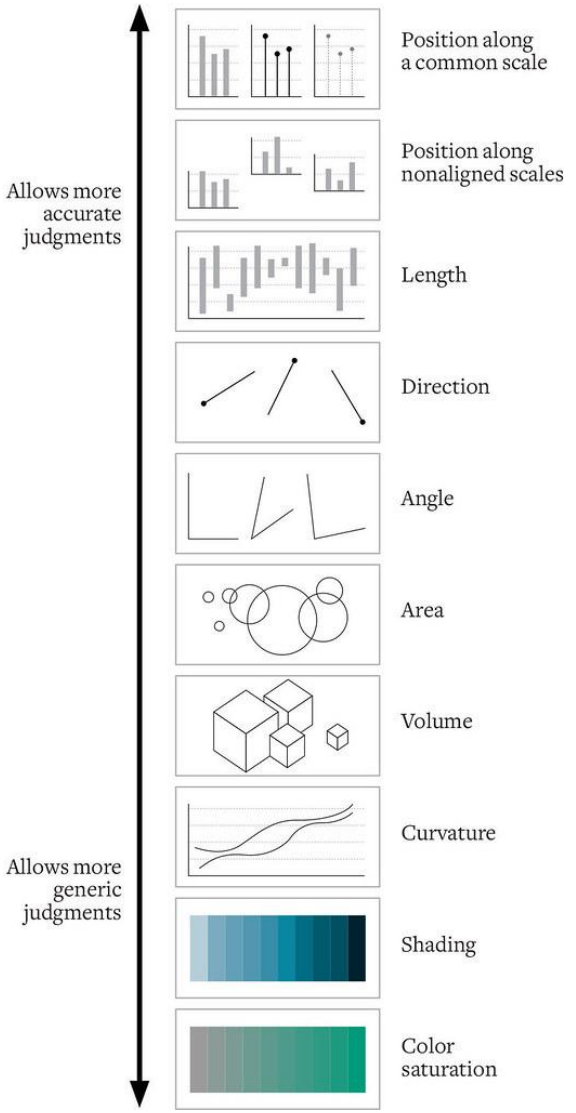
Choosing a chart type is choosing how much error you hand your reader.

THE GRAMMAR OF A CHART

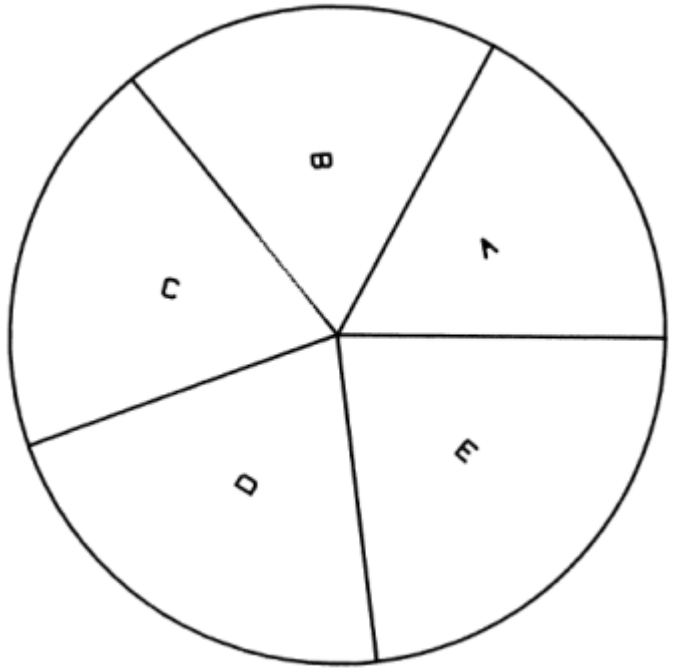
Cleveland and McGill's accuracy ladder

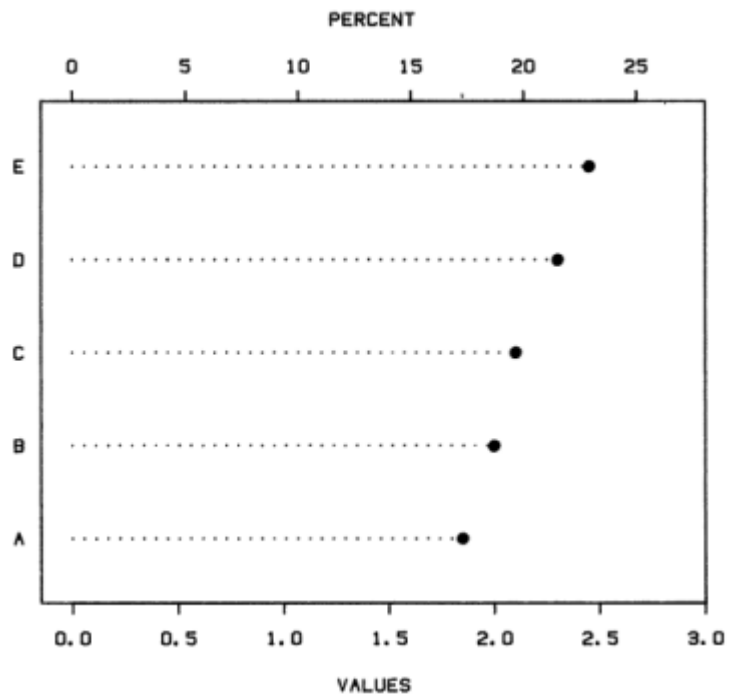
Ranked from most accurately decoded to least — an experimental result, not an opinion

Rank	Channel	Charts that use it	Error
1	Position, common scale	bar, dot plot, scatter, boxplot	lowest
2	Position, non-aligned scales	small multiples, facets	low
3	Length	stacked bar segments, floating bars	moderate
4	Angle or slope	pie, donut	high
5	Area	bubble, treemap, 3D columns	very high
6	Colour, shading	heatmap, choropleth map	highest



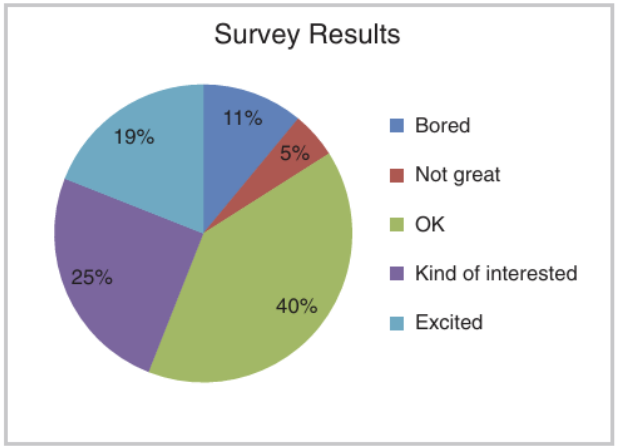
Move every quantity you want compared as high up this ladder as you can.

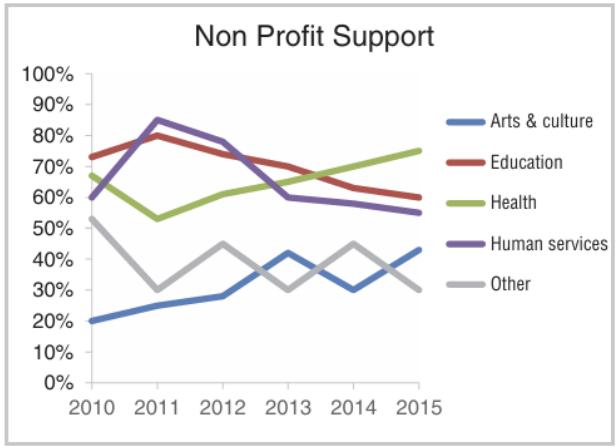

B
A
C
D
E

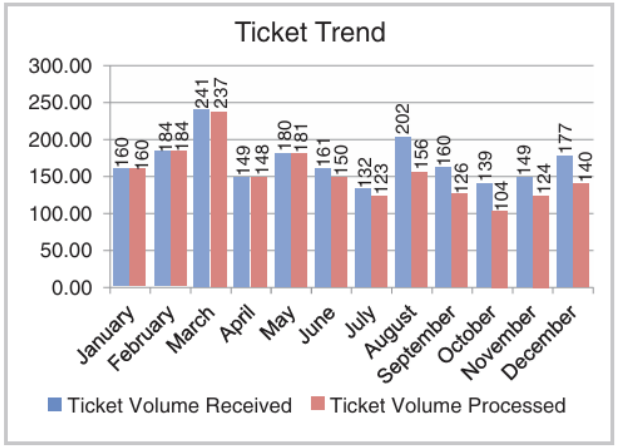

PERCENT
0
5
10
15
20
25
E
D
C
B
A
0.0
0.5
1.0
1.5
2.0
2.5
3.0
VALUES

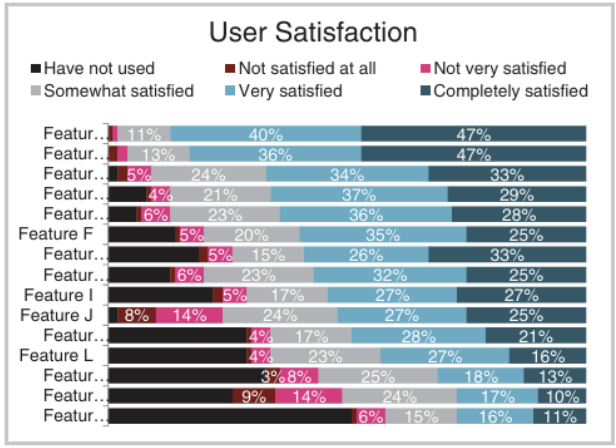
THE GRAMMAR OF A CHART

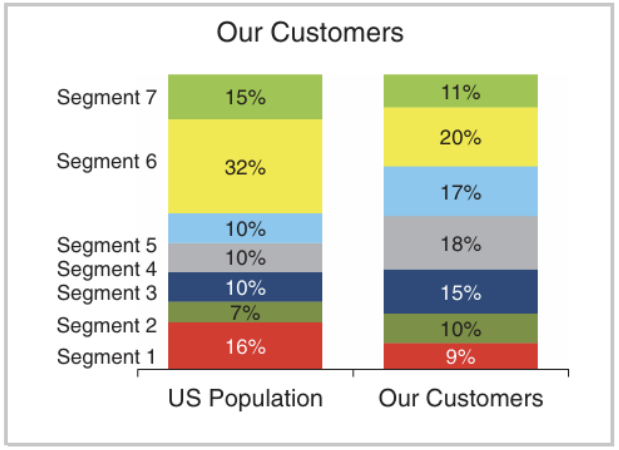
Not every chart is a “Good chart”

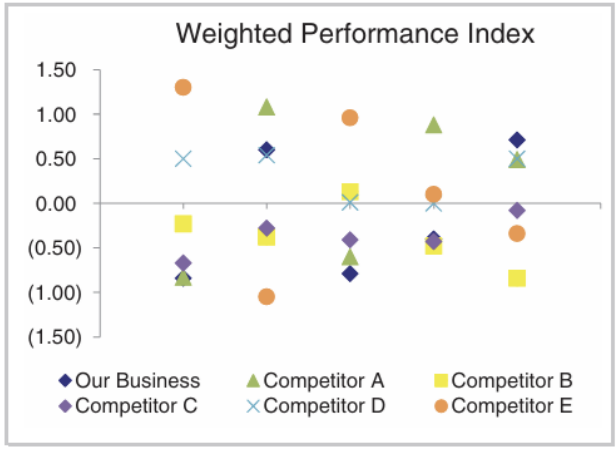












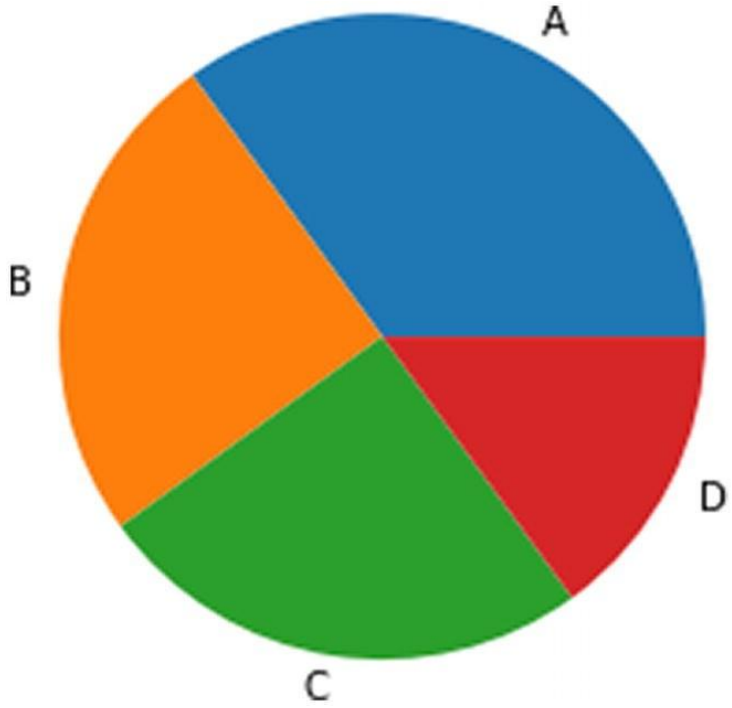
THE GRAMMAR OF A CHART

The pie chart test

Five market shares. Answer from the pie alone, before we show you the bars.

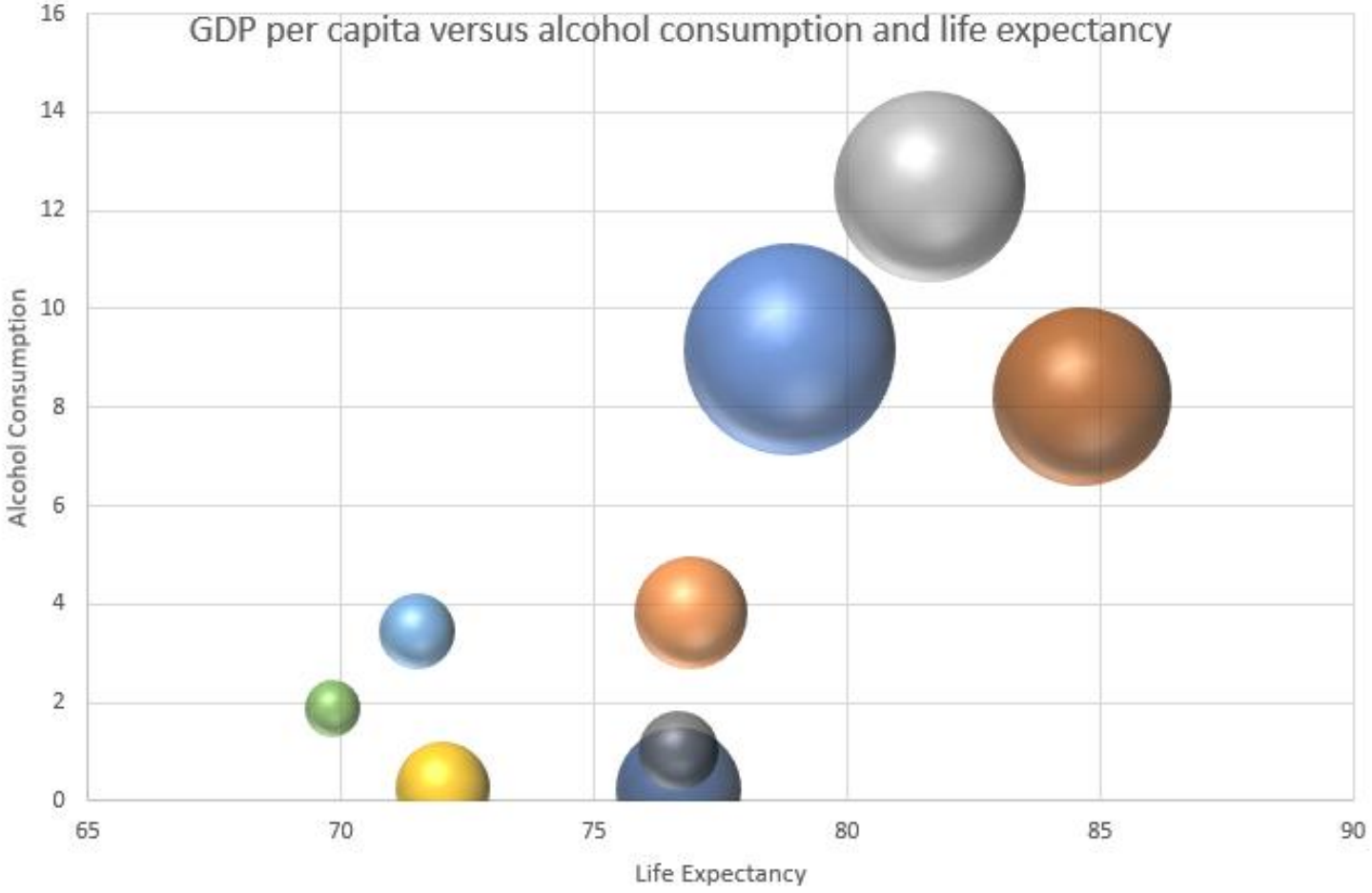
ANSWER FROM THE PIE ALONE

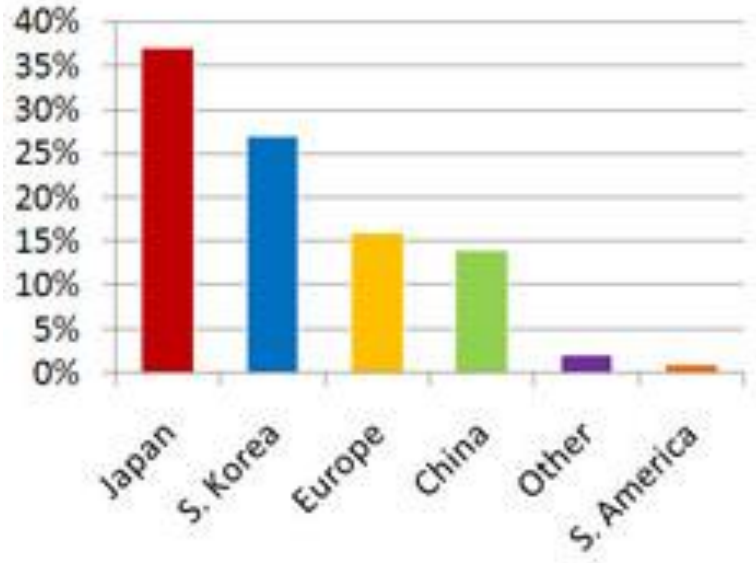
- 1 · Rank the 4 segments, largest to smallest
- 2 · Is segment C bigger or smaller than segment B?
- 3 · Is A 10 percent bigger than B, or 20 percent?

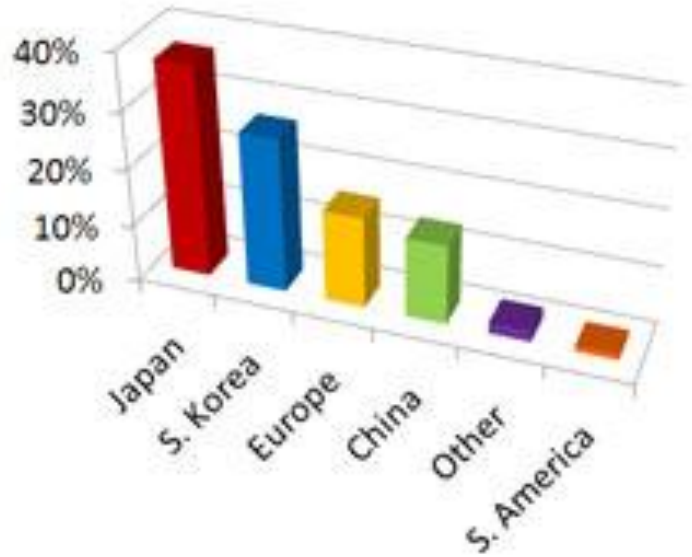


Which of the three questions could you actually answer?

Area, volume, and the 3D column







THE GRAMMAR OF A CHART

Area, volume, and the 3D column

ENCODING

Three ways to inflate a gap without changing a number

Map a value to a bubble's radius instead of its area, and doubling the value quadruples the ink. Draw a column in 3D and its apparent volume rises with the cube of its height.

READING

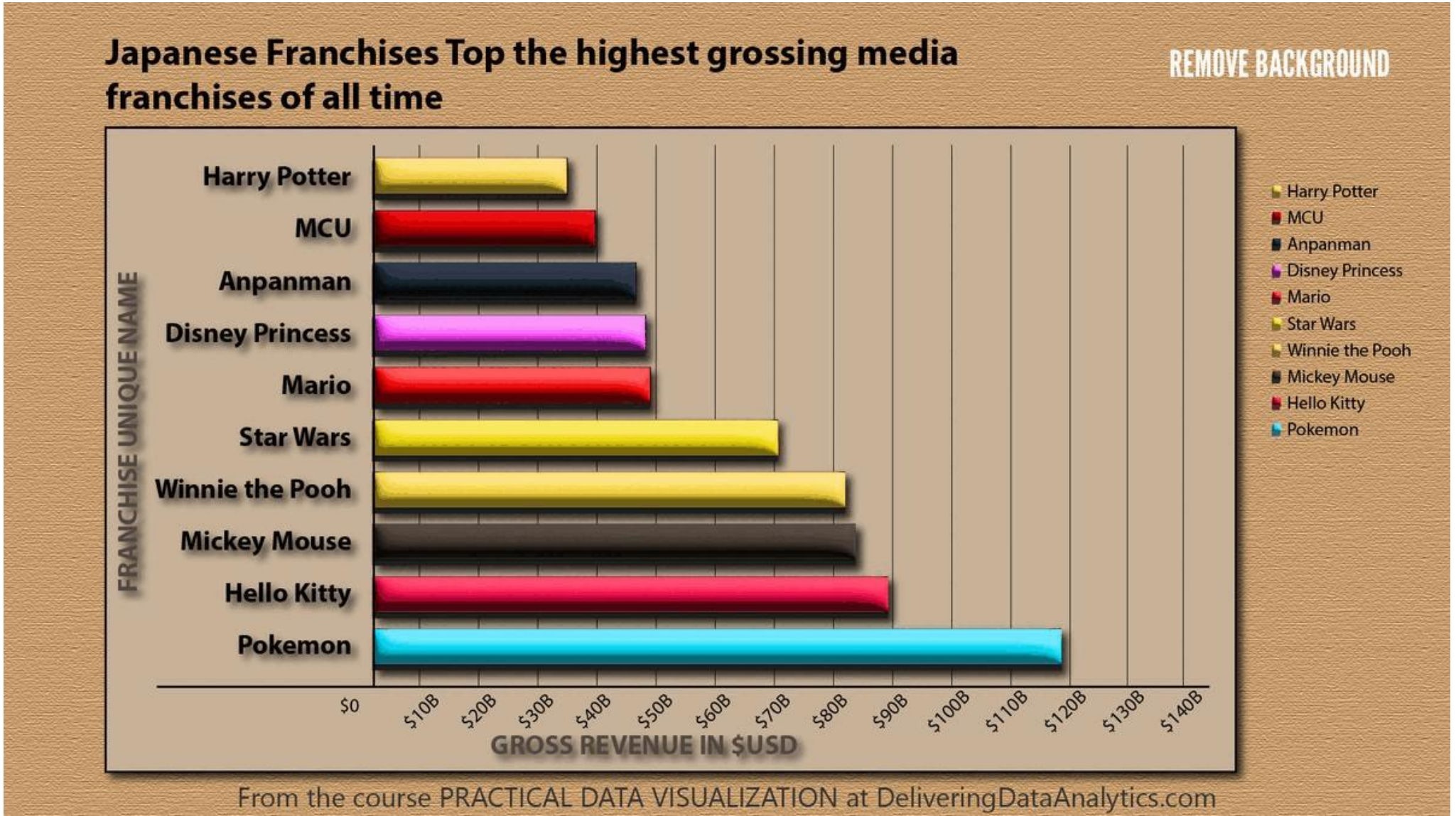
The reader interprets from the real value of the area, volume

A two-fold difference reads as four-fold or eight-fold. And because every underlying number is correct, the chart survives every fact-check. Nothing in it is false.

The fix is one line in a chart dialog: scale by area, remove 3D. This is not a budget problem.

Data-ink ratio: what to delete, in order

Everything on a chart is either data or noise. There is no third category.



THE GRAMMAR OF A CHART

Data-ink ratio: what to delete, in order

Everything on a chart is either data or noise. There is no third category.

STEP 1

Background and border

Neither came from your data. Delete both first.

STEP 2

Shadow, gradient, 3D

Decoration that actively distorts what you drew.

STEP 3

Gridlines

Keep one direction at most, thin and light grey.

STEP 4

Legend for one series

Label the series directly on the chart instead.

STEP 5

Then re-read the chart

Can you still answer the question it was drawn for? Then stop.

Six deletions, no information lost. But the chart is now much less complicated

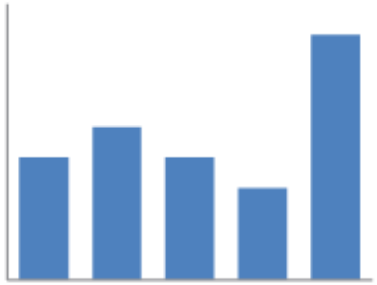
DIFFERENT TYPE OF CHARTS

91%

Simple text



Scatterplot



Vertical bar



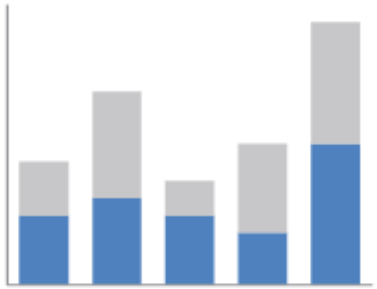
Horizontal bar

	A	B	C
Category 1	15%	22%	42%
Category 2	40%	36%	20%
Category 3	35%	17%	34%
Category 4	30%	29%	26%
Category 5	55%	30%	58%
Category 6	11%	25%	49%

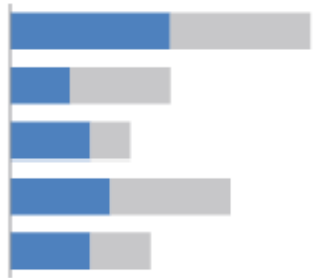
Table



Line



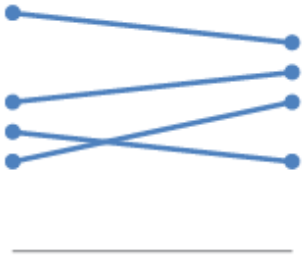
Stacked vertical bar



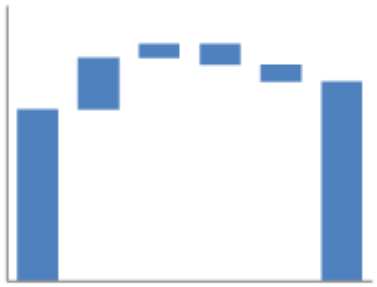
Stacked horizontal bar

	A	B	C
Category 1	15%	22%	42%
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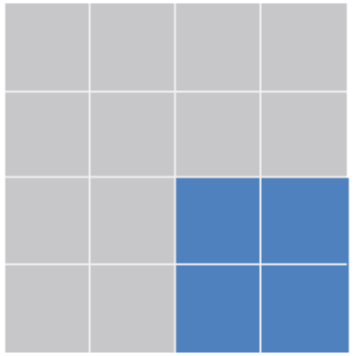
Heatmap



Slopegraph



Waterfall



Square area

DIFFERENT TYPE OF CHARTS

Build four figures on VN-LISTED

Latest-year cross-section — the same slice you described in Session 2

Build four figures in your own file: the log pair, a boxplot by industry, a scatter, and a correlation heatmap.

TIME

40 minutes, alongside the demo. Excel leads; Python does what Excel cannot.

FIGURES

Four figures and four sentences — one per figure, written the moment you finish it.

REQUIREMENTS

Each sentence completes: 'This shows ___, which the Session 2 table did not show.'

A figure with no sentence does not count as submitted. Bring the file to Session 5.

DIFFERENT TYPE OF CHARTS

The Barchart – Standard for Comparison

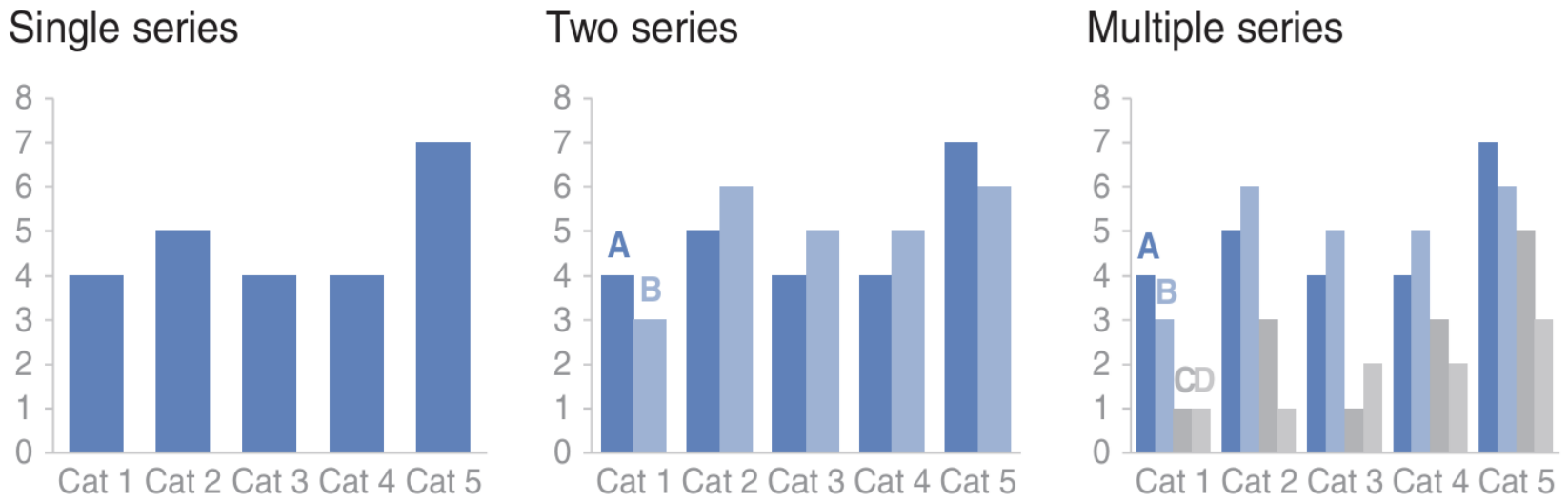
Barchart is useful for Comparison

- **Horizontal Bar:** Use when labels are long
- **Vertical Bar:** Use when X-axis represents time
- **Stacked Bar**: compare totals across categories and subcomponent pieces within a category
- **Waterfall Bar**: Breakdown subcomponent in the **Stacked Bar**

Mandatory Rule: The Y-axis for Bar Charts **must start at 0**.

Truncating the axis is the most common way to accidentally (or intentionally) lie with data.

Sort: If not ordered by time, sort from high to low

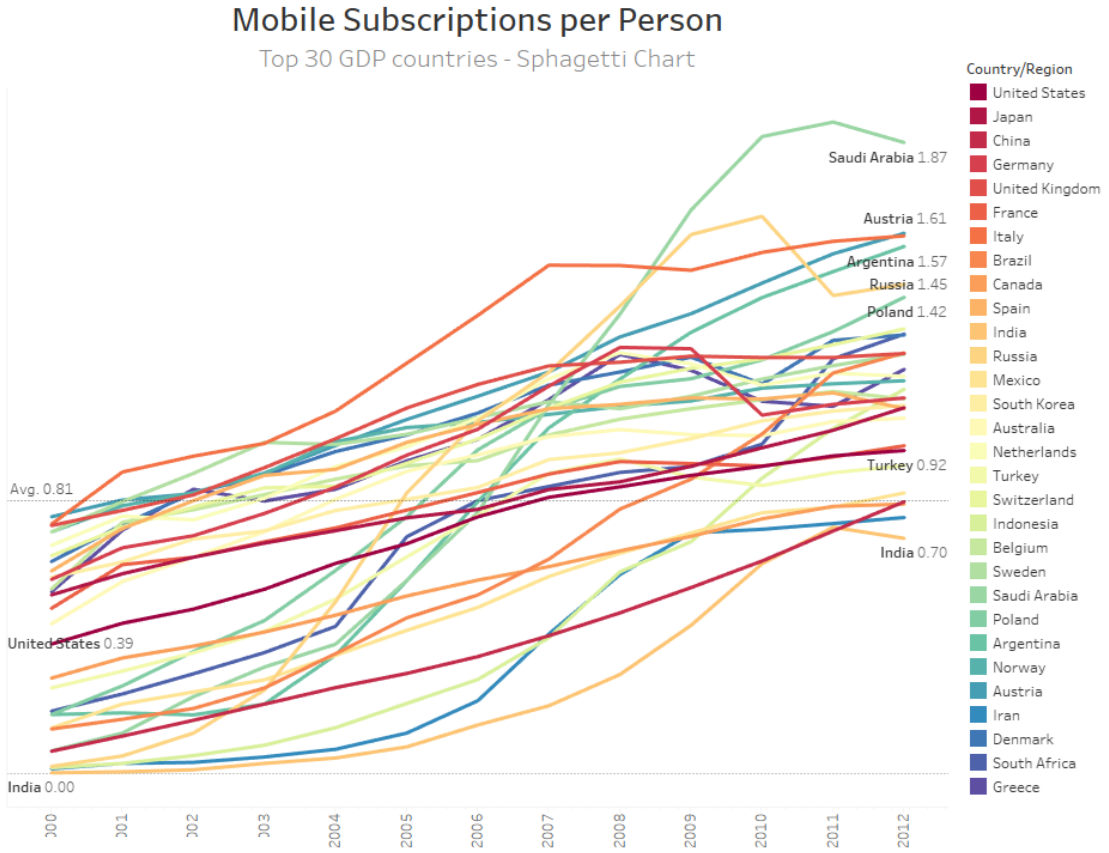


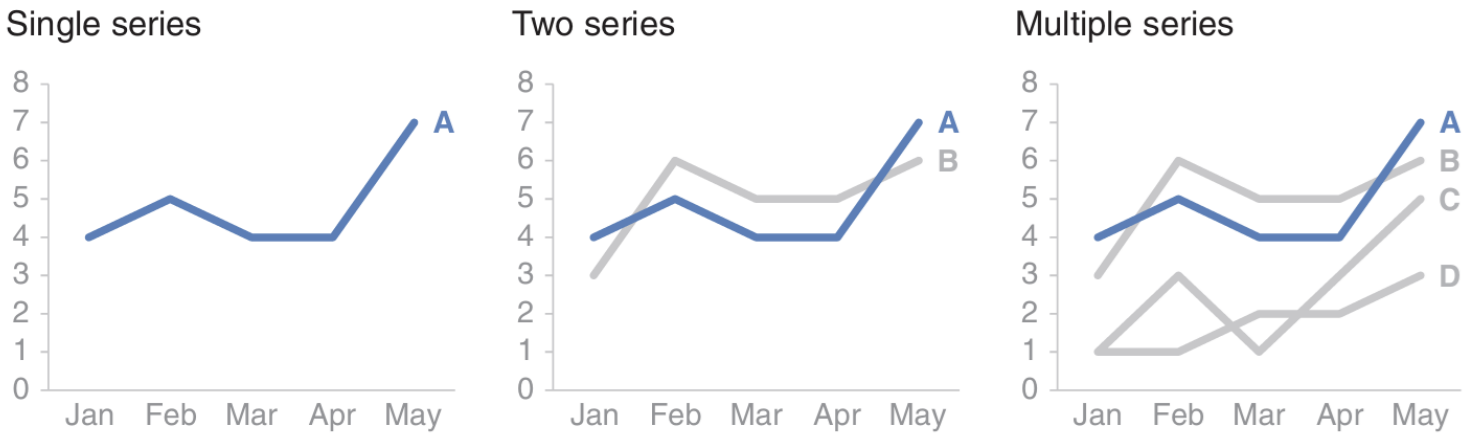
DIFFERENT TYPE OF CHARTS

The Line Graph – Tracking Continuity

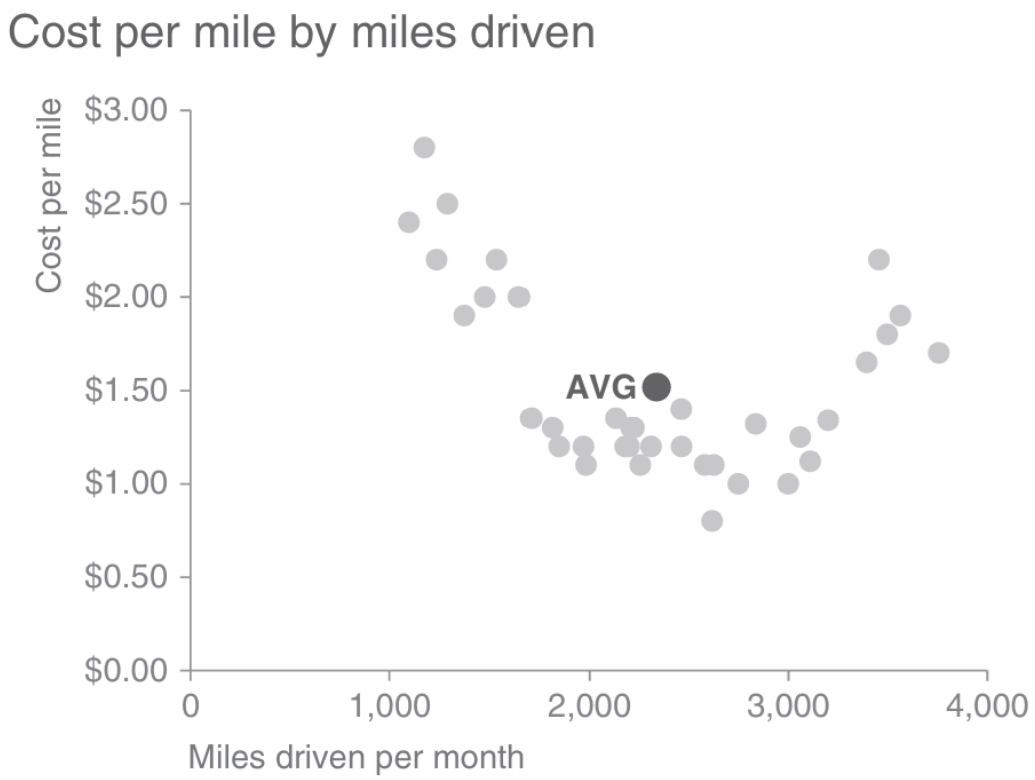
Lines imply continuity. They suggest that values between measured points exist and are changing.

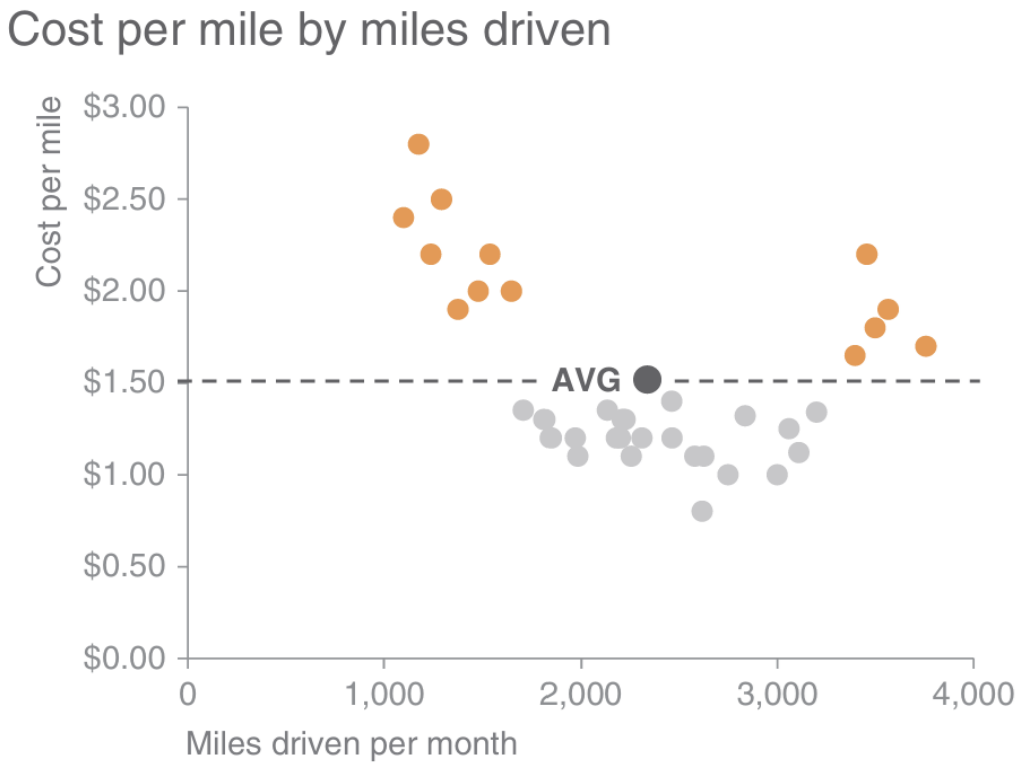
- Default for time-series data (Jan → Dec)
- Emphasize on the changes along the timeline
- Avoid "Spaghetti Charts": 15 lines in 15 colors is unreadable.
- Highlight the story line; grey out the rest.





The scatter plot, and what to check before you believe it





DIFFERENT TYPE OF CHARTS

The scatter plot, and what to check before you believe it

OVERPLOTTING

Dots land on top of each other. The dense centre may hold ten times more firms than it appears to. Fix with transparency or smaller markers.

ONE CLOUD OR TWO

Clusters almost always mean a third variable is doing the work. Colour by industry and look again before you read the overall shape.

WHAT DRIVES THE LINE

Cover the far right of the plot with your hand. Does the trendline still look like that? If not, a few firms are writing your conclusion.

LOG OR RAW

Two right-skewed variables give you a dense corner and a few distant dots. Log both axes first, then look at it again.

Today: do these move together?

DIFFERENT TYPE OF CHARTS

The histogram, and the choice hidden inside it

ENCODING

Two steps, and only the second one is arithmetic

First, cut the range into intervals — the bins. That decision comes from a human or from a default setting, never from your data. Second, count what falls in each bin and draw the count as a height.

READING

The first picture of a column's shape

- Where the mass of the data actually sits
- One peak or two — two peaks usually means two populations mixed
- How long the right tail runs
- Whether the mean sits inside the bulk, or alone out in the tail

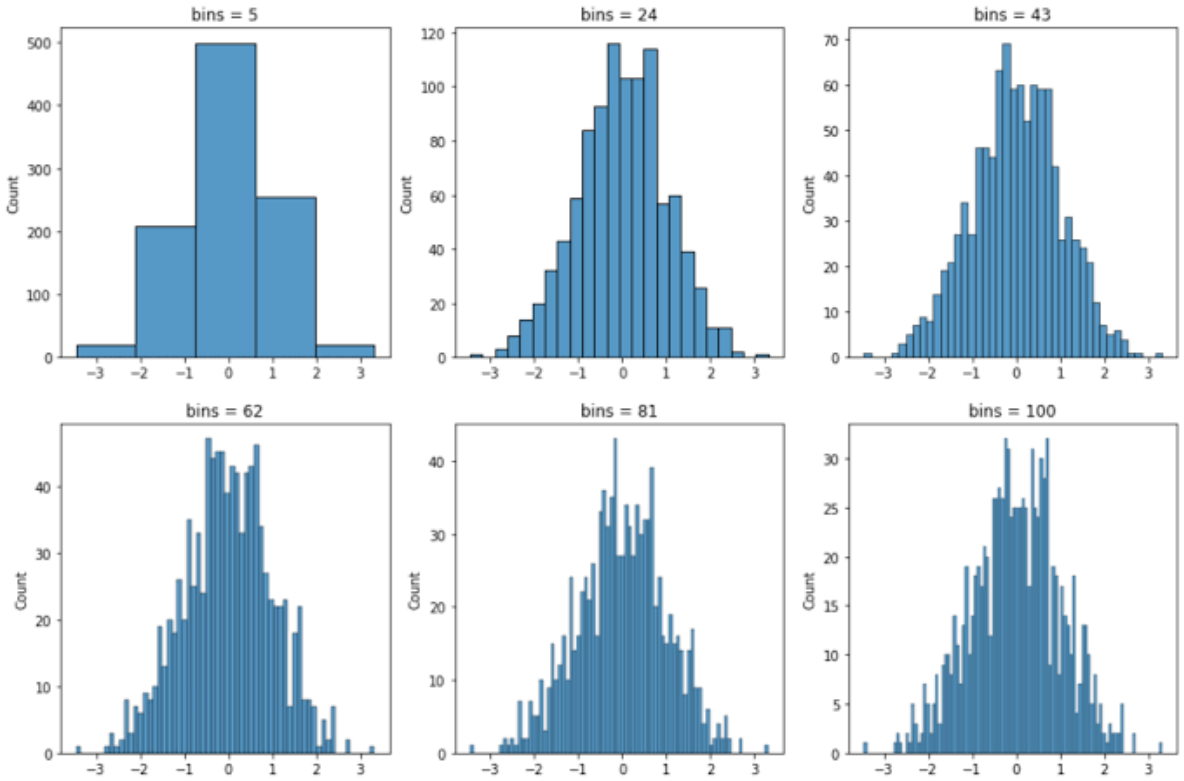
Step one is an editorial choice of yours. Declare it.

DIFFERENT TYPE OF CHARTS

How many bins are there?

READING

- Too few bins — smooth and symmetric; the right tail vanishes
- Too many bins — every gap in the data becomes a gap in the distribution
- Choose by stability: nudge the width 20%. What vanishes was never there



Which of the these is the most suitable histogram of this variable?

DIFFERENT TYPE OF CHARTS

The log pair

The single most important picture of this session

ENCODING

log turns multiplicative distance into additive

On the raw axis, 10 to 100 is a gap of 90 and 100 to 1,000 is a gap of 900. On the log axis both gaps are the same size, because both are 'times ten'. No observation was dropped. The ruler changed.

READING

The outliers were never outliers

- Raw scale — everything piled at the left wall, a few firms far right
- Log scale — the pile spreads into a near-symmetric shape
- Those right-hand firms are the top of a right-skewed distribution
- Session 5 draws a curve on this.

A right-skewed variable on a raw axis is a chart of its own outliers.

DIFFERENT TYPE OF CHARTS

The boxplot: five numbers you already computed

Every mark on this chart comes from your Session 2 table

BOX EDGES

Q1 and Q3. The width of the box is the IQR.

LINE INSIDE THE BOX

The median — not the mean. The mean is not drawn anywhere.

WHISKERS

The furthest observations still within 1.5 x IQR of the box.

DOTS BEYOND THE WHISKERS

The 1.5 x IQR flags from Session 2. Candidates, not errors.

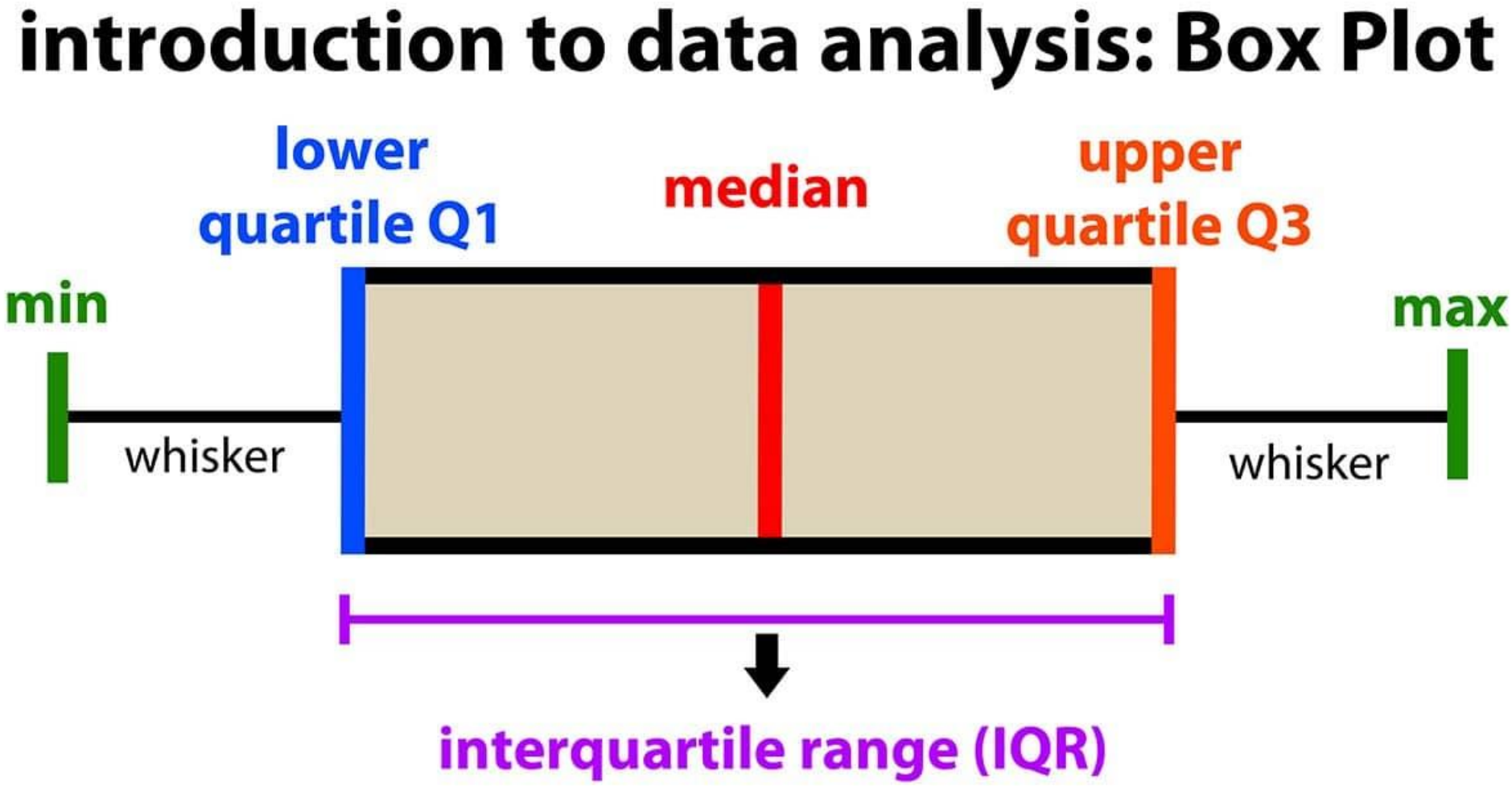
WHAT IT CANNOT SHOW

A second peak. A bimodal and a flat column give identical boxes.

If shape matters, put a histogram beside the box.

The boxplot: five numbers you already computed

Every mark on this chart comes from your Session 2 table



DIFFERENT TYPE OF CHARTS

The correlation heatmap: read the colour, not the number

The formula arrives in Session 7. Today you only learn to read the picture.

01

DARK CELLS

Pairs worth a scatter plot. This is a shortlist, not a finding.

02

PALE CELLS

No straight-line relationship. A strong curve also shows up pale.

03

BLOCKS OF DARK CELLS

Several variables measuring one thing. Session 13 calls this multicollinearity.

Colour is the worst channel on the ladder. A heatmap tells you which scatter to draw next — it is never the evidence.

DIFFERENT TYPE OF CHARTS

Putting a third variable on a flat page

Channel	Works when	Breaks when
Colour by category	Up to 3 categories	The legend becomes a lookup task
Size by quantity	Rarely, if ever	It inherits the area problem
Small multiples	Many groups, shared axes	A small group goes too sparse

Up to 3 categories, colour. More than that, facet — and the panels must share axes.

Workshop: diagnose and prescribe

One published chart each group, plus the question it was drawn to answer.

Four questions per card: What does it want you to believe? Which encoding does that? Which chart is correct? Does the conclusion survive?

TIME

5 min brief · 12 min group work · 18 min presenting, 3 min each · 10 min consolidation

FIGURES

One named channel, one named chart type, one hand-drawn honest version per card.

REQUIREMENT

Question 2 must name a channel from the ladder. 'Bad design' is not an answer.

Sketch the honest version on paper, by hand. No laptops for this part.

WORKSHOP — DIAGNOSE & PRESCRIBE

Reference table: question first, chart second

A tool for the workshop, not a list to memorise

The question you are asking	Use	Do not use
Shape of one numeric variable?	histogram, density, boxplot	pie, single bar
Groups compared on a numeric variable?	boxplot by group, dot plot	bar of means with no n
Groups compared on a count or total?	bar chart, baseline at zero	pie beyond 3 segments
Do two numeric variables move together?	scatter plot	two lines on a dual axis
How does something change over time?	line, x-axis chronological	bars reordered by value
Which of many pairs are worth a look?	heatmap, as screening only	the heatmap as a result
What share of a whole?	stacked bar, or a table	pie, donut, 3D pie

Read the left column first. Always. A chart is the answer to a question, so you must have the question.

WORKSHOP — DIAGNOSE & PRESCRIBE

Topic: Choose alongside of these

Every number in all three charts is correct. Find the encoding decision.

1

A tax-rate change, on cable news

Two bars, a few points apart. The y-axis starts just below the lower bar.

2

Firearm deaths after a law changed

A filled area over time. The y-axis runs downward: larger values sit lower.

3

COVID-19 cases by county

Grouped bars. Counties re-sorted inside each date, and the dates are out of order.

4

Revenue growth, investor deck

Three 3D columns, three years. The y-axis starts just below the first column.

5

Revenue and share price together

Two lines, two y-axes. The ranges were chosen so the lines nearly overlap.

6

Average ROA by industry

Eight bars. No n on any bar, no spread. Two industries hold very few firms.

WORKSHOP — DIAGNOSE & PRESCRIBE

Groups 4-6: the cases you will meet at work

Reconstructed from patterns common in annual reports and investor decks

4

Revenue growth, investor deck

Three 3D columns, three years. The y-axis starts just below the first column.

5

Revenue and share price together

Two lines, two y-axes. The ranges were chosen so the lines nearly overlap.

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Average ROA by industry

Eight bars. No n on any bar, no spread. Two industries hold very few firms.

Group 6 — you drew the honest version of this chart forty minutes ago, on slide 17.

WORKSHOP — DIAGNOSE & PRESCRIBE

The catalogue you just built

Every row came out of a group presentation. Now check it for gaps.

The trick	The channel it exploits	The honest version
Truncated axis	position, baseline moved	baseline at zero for bars
Inverted axis	position, direction reversed	conventional direction
Re-sorted x-axis	position, ordering destroyed	strict chronological order
Dual axis	position, two incomparable scales	two charts, or index both to 100
Area for a 1D quantity	area instead of length	length on a common baseline
Cherry-picked window	position, domain restricted	show the full available series
Mean with no n, no spread	one number for a distribution	boxplot, or bars with n

Is there an entire category of deception missing from what the six groups found?

WORKSHOP — DIAGNOSE & PRESCRIBE

Four questions before you draw anything

The routine to run every time, for the rest of this course

STEP 1

What is the question?

Write it with a subject and a verb. 'Revenue by industry' is not a question.

STEP 2

Which variables answer it?

How many, and numeric or categorical? This alone rules out most charts.

STEP 3

Which channel can I use?

Go as high up the accuracy ladder as your data allows. Position first.

STEP 4

How could this mislead?

Run your own chart against the catalogue before anyone else does.

Steps 1 to 3 make the claim. Step 4 is whether you can defend it.

PUBLICATION STANDARD

A figure that meets publication standard

Every figure you submit from today onward is marked against these rows

Element	Requirement	The usual failure
Title	States the conclusion, not the contents	'Figure 1: Revenue distribution'
Axis labels	Variable name and unit	'Revenue', no unit anywhere
Scale note	If transformed, say so on the axis	log applied silently
Sample size	n = ___, visible on the figure	omitted
Source	Dataset, coverage, extraction date	omitted
Treatment	Rows dropped? Say which, and why	silent deletion
Colour	Legible in greyscale and to a colour-blind reader	red versus green

Everything else — 3D, gradients, shadows, redundant legends — delete it.

SUMMARY

What today changed, and what it hands forward

1. THE MECHANISM

A statistic is a sum over rows, so it cannot remember rows. Structural, and no amount of care removes it.

2. THE LADDER

The eye decodes each channel with a fixed accuracy. Chart choice is error management, not taste.

3. YOUR DECISIONS

Every misleading chart today was factually correct. Baseline, ordering and sample size are yours.

Session 4 — we stop describing what the data is, and start reasoning about what it could have been.